1. Program to read Day, Month, and Year. Then display the corresponding date in DD-MM-YYYY format. Also, Check whether the date is from a leap or not.

```
# Read day, month, and year
day = int(input("Enter the day (1-31): "))
month = int(input("Enter the month (1-12): "))
year = int(input("Enter the year (e.g., 2024): "))

# Display the date in DD-MM-YYYY format
print(f"Entered date: {day:02d}-{month:02d}-{year}")

# Check if the year is a leap year
if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
    print(f"The year {year} is a leap year.")
else:
    print(f"The year {year} is not a leap year.")
```

2. Program to read three numbers and find the biggest. Let ‘n’ is the biggest number, compute n+nn+nnn. Assume the same ‘n’ is the radius of a circle and find its Area and Perimeter. Find the volume of a sphere with a radius ‘n’.

```
# Approximation of pi
pi = 3.1416

# Read three numbers from the user
num1 = int(input("Enter the first number: "))
num2 = int(input("Enter the second number: "))
num3 = int(input("Enter the third number: "))

# Find the biggest number
if num1 >= num2 and num1 >= num3:
    n = num1
elif num2 >= num1 and num2 >= num3:
    n = num2
else:
    n = num3

print(f"The biggest number is: {n}")
# Compute n + nn + nnn
nn = n**2   # Concatenate n with itself
nnn = n**3  # Concatenate n three times
```

```
result = n + nn + nnn
print(f"n + nn + nnn = {n} + {nn} + {nnn} = {result}")

# Assume n is the radius of a circle
radius = n

# Calculate Area and Perimeter of the circle
area = pi * radius ** 2
perimeter = 2 * pi * radius

print(f"\nFor a circle with radius {radius}:")
print(f"Area = {area:.2f}")
print(f"Perimeter (Circumference) = {perimeter:.2f}")
```

3. Create a list of colors (color list1) from comma-separated color names entered by the user. Display the first and last colors. Read another color list and print out all colors from color-list1 not contained in color-list2. Create another List from color list 1 by assigning an integer value for each color.

```
color=input("ENTER COLORS SEPERATED BY COMAS:")
color_list1=color.split(',')
print(color_list1)

print("FIRST COLOR:", color_list1[0])
print("LAST COLOR:", color_list1[-1])
color_list2 = ["Red", "Orange", "Black", "White"]

print("color-list1 not contained in color-list2 are : ")
diff=list(set(color_list1)- set(color_list2))
print(diff)
color_int_list=[]
for color in color_list1:
  color_int_list.append(len(color))

print(f"List of integers corresponding to each color: {color_int_list}")
```

4. Create a Dictionary. Sort the dictionary in ascending and descending order. Create another dictionary and Merge these two dictionaries.

```
dict1={2:"orange",3 : "banana",1 :"apple"}
print(f"Dictionary 1: {dict1}")
l=list(dict1.items())
l.sort()
print('Ascending order is : ', l)
l=list(dict1.items())
l.sort(reverse=True)
print('Descending order is : ', l)

dict2={4 :"plum",5 :"cherry"}
print(f"Dictionary 1: {dict2}")
dict1.update(dict2)
print(f"Dictionary after merging:{dict1}")
```